

HAY, Australia

WMO: 947020

Lat: 34.5411S

Lon: 144.8344E

Elev: 303

StdP: 14.54

Time Zone: 10.00 (AUS)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	31.5	33.8	19.4	14.8	75.1	22.9	17.4	72.4	26.4	56.4	24.4	57.4	4.3	250	0.550

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	27.5	104.1	67.2	100.1	66.2	96.4	65.5	73.7	86.5	71.2	86.0	69.5	85.1	13.6	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
71.1	116.1	75.3	67.7	103.0	74.8	64.8	92.8	74.1	37.5	87.5	35.2	85.9	33.7	84.8	98.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.4	22.5	20.1	27.4	109.7	2.6	4.4	25.5	112.9	23.9	115.5	22.5	118.0	20.6	121.2		
(5)			26.8	79.1	2.3	9.0	25.1	85.6	23.8	90.8	22.4	95.9	20.7	102.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	62.9	78.3	76.9	70.9	62.5	55.3	50.2	49.0	50.9	56.2	62.6	69.2	74.0		
	DBStd	12.35	8.12	7.82	7.69	6.74	5.44	4.65	4.30	5.05	6.56	7.62	8.53	8.20		
	HDD50	195	0	0	0	0	11	54	73	46	10	1	0	0		
	HDD65	2322	2	5	27	125	305	445	497	438	279	142	45	12		
	CDD50	4912	878	754	647	377	175	58	41	74	195	391	577	745		
	CDD65	1563	414	339	209	52	3	0	0	1	14	68	172	291		
	CDH74	21205	5815	4331	2482	612	33	0	0	18	244	1063	2436	4171		
	CDH80	11115	3424	2373	1156	180	1	0	0	2	63	426	1234	2256		
(14) Wind	WSAvg	10.2	11.0	10.7	10.2	8.7	8.9	8.4	9.5	10.1	10.8	11.0	11.3	11.4		
(15) Precipitation	PrecAvg	15.7	1.5	1.1	1.3	1.1	1.6	1.2	1.3	1.3	1.5	1.5	0.9	1.2		
	PrecMax	33.0	7.5	8.0	7.2	5.9	5.3	2.9	3.6	2.8	3.9	5.9	5.6	4.5		
	PrecMin	6.5	0.0	0.0	0.0	0.0	0.0	0.2	0.1	0.0	0.2	0.0	0.0	0.0		
	PrecStd	5.8	1.6	1.7	1.7	1.2	1.2	0.8	0.9	0.7	1.0	1.3	0.9	1.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	110.1	107.8	101.2	90.2	78.0	69.4	68.6	76.4	86.4	95.5	102.8	106.0		
		MCWB	67.7	67.7	66.7	62.1	58.5	56.6	53.7	55.0	58.4	60.5	65.1	65.5		
	2%	DB	105.3	101.5	95.1	84.9	73.4	65.1	63.8	70.1	80.7	89.7	97.1	100.9		
		MCWB	67.6	67.9	65.4	60.8	57.2	54.8	52.4	53.6	57.5	60.1	64.2	65.1		
	5%	DB	100.6	97.0	90.7	81.1	69.5	62.0	60.7	65.7	75.8	84.6	92.3	96.5		
		MCWB	66.8	67.1	64.6	60.1	56.5	53.3	51.7	52.0	56.5	59.1	63.2	64.8		
	10%	DB	95.9	93.0	86.5	76.9	65.9	59.4	58.1	62.1	70.6	79.8	87.2	92.1		
		MCWB	65.9	66.1	63.2	58.9	55.3	52.4	50.6	51.0	54.7	57.7	61.0	63.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	76.4	77.0	72.3	66.5	63.5	59.3	58.7	59.0	62.9	67.0	77.5	87.0		
		MCDB	84.9	84.5	83.7	77.2	67.2	64.9	61.8	68.5	74.1	80.1	77.5	87.7		
	2%	WB	72.4	73.4	70.0	64.4	60.1	56.7	55.3	56.1	60.6	63.8	68.3	71.1		
		MCDB	88.8	87.8	81.9	74.3	67.8	62.1	59.7	64.8	72.7	77.5	83.7	81.5		
	5%	WB	70.5	71.3	67.8	62.5	58.3	55.0	52.9	53.9	58.4	61.8	66.0	68.1		
		MCDB	87.9	86.1	81.8	74.1	66.0	59.9	58.5	62.5	70.4	77.3	81.7	83.8		
	10%	WB	68.8	69.4	65.7	60.7	56.6	53.3	51.3	52.1	56.1	59.7	64.0	65.8		
		MCDB	87.9	84.9	81.1	74.6	64.5	58.4	56.9	60.3	68.7	76.9	80.5	86.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	27.5	25.6	24.9	24.6	21.4	19.1	19.1	22.1	25.1	27.2	26.9	27.4		
		MCDBR	32.0	29.7	28.2	28.8	24.8	22.8	22.4	27.7	30.9	32.8	31.5	30.8		
		MCWBR	10.4	10.2	10.5	12.7	12.6	13.6	13.4	14.5	13.3	12.8	11.0	11.1		
		MCDBR	24.9	22.7	21.8	23.2	19.3	18.2	18.9	22.9	25.9	26.4	24.7	23.7		
	5% WB	MCWBR	11.2	10.3	9.8	11.8	10.9	11.7	12.3	13.5	13.2	12.8	10.8	12.7		
(40) Clear-Sky Solar Irradiance	taub		0.336	0.338	0.317	0.305	0.295	0.288	0.288	0.295	0.325	0.327	0.342	0.333		
	taud		2.490	2.503	2.550	2.551	2.553	2.573	2.567	2.536	2.442	2.458	2.448	2.482		
	Ebn at Noon		318	310	305	291	277	271	277	290	297	310	313	319		
	Edh at Noon		36	35	31	27	24	22	23	27	34	36	38	37		
(44) All-Sky Solar Radiation	RadAvg		2501	2222	1866	1381	990	792	865	1176	1618	2047	2291	2524		
	RadStd		133	113	100	60	49	55	47	72	102	122	129	108		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)		+0.77	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+283	+227

Nomenclature: See separate page